

Assignment: Product Inventory Analysis for Apex Global Warehouse

Overview

In this assignment, you will apply your SQL and data analysis skills to a real-world dataset. You will act as a Data Analyst for a large warehouse business, using BigQuery to answer critical business questions that will drive the company's sales strategy.

Business Questions & SQL Answers

1. Quick Stock Check

Question: Management wants a quick look at our inventory. Show the *product_name*, *category*, and *actual_price* for the first 10 products.

Sandbox
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Run
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```

1 Select product_name,category,actual_price
2 From tkh-project-485015.warehouse_analysis.amazon_products
3 LIMIT 10

```

✓ Query completed
Using on-demand processing quota

Query results
+ Create conversation
📄 ▼
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Job information
Results
Visualization
JSON
Execution details
>

Row	product_name ▼	category ▼	actual_price ▼
1	Khaitan ORFin Fan heater for Ho...	Home&Kitchen Heating,Coolin g&AirQuality RoomHeaters Fa nHeaters	2495.0
2	Personal Size Blender, Portable Blender, Battery Powered USB Blender, with	Home&Kitchen Kitchen&Home Appliances SmallKitchenAppli ances HandBlenders	1499.0

Results per page:
50 ▼
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2. Identifying High-Value Items

Question: Find all products where the *actual_price* is greater than 5,000. Show the name and the price.

The screenshot shows the Google Cloud BigQuery console interface. The left sidebar contains navigation options: Overview, Studio, Agents, Pipelines & Integration, Data transfers, Dataform, Scheduled queries, Scheduling, Governance, Sharing (Analytics Hub), Policy tags, Metadata curation, Administration, Monitoring, Jobs explorer, Capacity management, Partner Center, Settings, and Release Notes. The main area displays a query titled "Identifying High-Value Items" with the following SQL code:

```
1 SELECT product_name, actual_price
2 FROM tkh-project-485015.warehouse_analysis.amazon_products
3 WHERE actual_price > 5000
```

Below the query editor, a message indicates "Query completed". The "Query results" section shows a table with two columns: "product_name" and "actual_price". The table contains two rows of data:

Row	product_name	actual_price
1	PTron Newly Launched Force X10 Bluetooth Calling Smartwatch with 1.7" Full Touch Color Display, Real Heart Rate Monitor, SpO2,	5999.0
2	PTron Newly Launched Force X10 Bluetooth Calling Smartwatch with 1.7" Full	5999.0

At the bottom right, it indicates "Results per page: 50" and "1 - 50 of 321".

3. The Discount Impact

Question: Calculate the total amount saved (Actual Price - Discounted Price) for each product. Call this new column *total_savings*. (is this AS)

The screenshot shows the Google Cloud BigQuery console interface. The left sidebar contains navigation options: Overview, Studio, Agents, Pipelines & Integration, Data transfers, Dataform, Scheduled queries, Scheduling, Governance, Sharing (Analytics Hub), Policy tags, Metadata curation, Administration, Monitoring, Jobs explorer, Capacity management, Partner Center, Settings, and Release Notes. The main area displays a query titled 'The Discount Impact' with the following SQL code:

```
1 SELECT product_name, actual_price - discounted_price AS total_savings
2 FROM tkh-project-485015.warehouse_analysis.amazon_products
```

Below the query, a message states: 'This query will process 209.67 KB when run. Using on-demand processing quota'. The 'Query results' section shows a table with the following data:

Row	product_name	total_savings
1	Khaitan ORFin Fan heater for H...	1196.0
2	Personal Size Blender, Portable Blender, Battery Powered USB Blender, with Four Blades, Mini Blender Travel Bottle for Juice, Shakes,	830.0
3	Green Tales Heat Seal Mini Food Sealer-Impulse Machine	139.0

At the bottom right, it indicates 'Results per page: 50' and '1 - 50 of 1465'.

4. Inventory Diversity

Question: How many unique products do we have in each category? Show the category and the count.

The screenshot shows the Google Cloud BigQuery console interface. The left sidebar contains navigation options: Overview, Studio, Agents, Pipelines & Integration, Governance, and Administration. The main area displays a query titled 'Inventory Diversity' with the following SQL code:

```
1 SELECT DISTINCT category, COUNT(category)
2 FROM tkh-project-485015.warehouse_analysis.amazon_products
3 GROUP BY category
```

Below the query, a message states: 'This query will process 107.17 KB when run. Using on-demand processing quota'. The 'Query results' section is active, showing a table with 3 rows and 2 columns: 'category' and 'count_1'.

Row	category	count_1
1	Home&Kitchen Heating,Cooling&AirQuality RoomHeaters FanHeaters	20
2	Home&Kitchen Kitchen&Home Appliances SmallKitchenAppliances HandBlenders	19
3	Home&Kitchen Kitchen&Home	7

At the bottom right, it indicates 'Results per page: 50' and '1 - 50 of 211'.

5. Customer Favorites

Question: We need a list of highly-rated products. Find products that have a *rating* higher than 4.5 and more than 10,000 *rating_count*.

Google Cloud console interface showing a BigQuery query execution. The query filters for products with a rating greater than 4.5 and a rating count greater than 10000. The results table shows two products: Logitech G402 Hyperion Fury and Logitech M331 Silent Plus Wireless Mouse.

```
1 SELECT product_id, product_name, rating, rating_count
2 FROM tkh-project-485015.warehouse_analysis.amazon_products
3 WHERE SAFE_CAST(rating AS float64) > 4.5 AND rating_count > 10000;
```

This query will process 222.42 KB when run.
Using on-demand processing quota Processing location: US

Query results

Row	product_id	product_name	rating	rating_count
1	B00NFD0ETQ	Logitech G402 Hyperion Fury USB Wired Gaming Mouse, 4,000 DPI, Lightweight, 8 Programmable Buttons, Compatible for PC/Mac -	4.6	10760
2	B01MQ2A86A	Logitech M331 Silent Plus Wireless Mouse, 2.4GHz with	4.6	12375

Results per page: 50 1 - 6 of 6

6. Discount Strategy Analysis

Question: What is the average *discount_percentage* for products in the 'Electronics' category?

The screenshot displays the Google Cloud BigQuery interface. On the left is a navigation menu with options like Overview, Studio, Agents, Pipelines & Integration, Data transfers, Dataform, Scheduled queries, Scheduling, Governance, Sharing (Analytics Hub), Policy tags, Metadata curation, Administration, Monitoring, Jobs explorer, Capacity management, Partner Center, Settings, and Release Notes. The main area shows a query titled "Discount Strategy Analysis" with the following SQL code:

```
1 SELECT AVG(discount_percentage), Category
2 FROM tkh-project-485015.warehouse_analysis.amazon_products
3 GROUP BY Category
4 HAVING Category LIKE '%Electronics%'
```

Below the query, a message states: "This query will process 118.61 KB when run. Using on-demand processing quota". The "Query results" section shows a table with 6 rows and 3 columns: Row, f0_ (discount percentage), and Category.

Row	f0_	Category
1	0.698157894736...	Electronics Wearable Technolog...
2	0.595102040816...	Electronics HomeTheater,TV&VI...
3	0.382857142857...	Electronics HomeTheater,TV&Video Televisions SmartTelevi...
4	0.582692307692...	Electronics Headphones,Earbud...
5	0.23	Electronics HomeTheater,TV&VI...
6	0.185555555555...	Electronics Mobles&Accessori...

At the bottom right, it indicates "Results per page: 50" and "1 - 50 of 65".

7. Cleaning Up the Ratings

Question: Some ratings are missing or corrupted (like the '|' character). Use a CASE statement to show the *product_name* and a new column that turns any rating of '|' into 0.0, otherwise keep the original rating.

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Untitled query Run Save Download Share Schedule Open in More

```

1 SELECT
2   product_name,
3   CASE
4     WHEN rating = '' THEN 0.0
5     ELSE SAFE_CAST (rating AS FLOAT64)
6   END AS cleaned_rating
7 FROM tkh-project-485015.warehouse_analysis.amazon_products
8
9

```

Query completed

Using on-demand processing quota

Query results + Create conversation Save results Open in

Job information Results Visualization JSON Execution details Execution graph

Row	product_name	cleaned_rating
1	Khaitan ORFin Fan heater for H...	2.0
2	Personal Size Blender, Portable Blender, Battery Powered USB Blender, with Four Blades, Mini Blender Travel Bottle for Juice, Shakes,	2.3
3	Green Tales Heat Seal Mini	2.6

Results per page: 50 1 - 50 of 1465

8. Pricing Brackets

Question: Create a column called *price_bracket* where:

- actual_price < 500 is 'Budget'
- actual_price between 500 and 2000 is 'Mid-Range'
- actual_price > 2000 is 'Premium'

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- Metadata curation

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- Settings **Preview**
- Release Notes

Pricing Brackets Run Share Schedule Open in More Save query Download

```

1 SELECT product_name, actual_price,
2 CASE
3   WHEN actual_price < 500 THEN 'Budget'
4   WHEN actual_price BETWEEN 500 AND 2000 THEN 'Mid-Range'
5   WHEN actual_price > 2000 THEN 'Premium'
6 END AS price_bracket
7 FROM tkh-project-485015.warehouse_analysis.amazon_products

```

✓ This query will process 198.22 KB when run.
Using on-demand processing quota

Query results + Create conversation Save results Open in

Job information	Results	Visualization	JSON	Execution details	Execution graph
Row	product_name	actual_price	price_bracket		
1	Khaitan ORFin Fan heater for H...	2495.0	Premium		
2	Personal Size Blender, Portable Blender, Battery Powered USB Blender, with Four Blades, Mini Blender Travel Bottle for Juice, Shakes,	1499.0	Mid-Range		
3	Green Tales Heat Seal Mini Food Sealer-impulse Machine	300.0	Budget		

Results per page: 50 1 - 50 of 1465

9. Missing Data Audit

Question: Identify any products where the *rating_count* is missing. This is crucial for our data integrity report.

The screenshot shows the Google Cloud BigQuery console interface. The left sidebar contains navigation links for Overview, Studio, Agents, Pipelines & Integration, Governance, and Administration. The main area displays a query titled "Missing Data Audit" with the following SQL code:

```
1 SELECT product_id, product_name, rating_count
2 FROM tkh-project-485015.warehouse_analysis.amazon_products
3 WHERE rating_count IS NULL
```

Below the query, a status message indicates: "This query will process 215.37 KB when run. Using on-demand processing quota". The "Query results" section shows a table with 2 rows and 4 columns: product_id, product_name, rating_count, and an empty column. The results are as follows:

Row	product_id	product_name	rating_count	
1	BOB94JPY2N	Amazon Brand - Solimo 65W Fast Charging Braided Type C to C Data Cable Suitable For All Supported Mobile Phones (1 Meter, Black)	null	
2	BOBQRJ3C47	REDTECH USB-C to Lightning Cable 3.3FT, [Apple MFI Certified] Lightning to Type C	null	

At the bottom right, it shows "Results per page: 50" and "1 - 2 of 2".

10. Competitive Categories

Question: Which categories have an average rating above 4.0? Only show categories that have more than 5 products.

Google Cloud

TKH Project

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Sandbox

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```
1 SELECT category, AVG(SAFE_CAST(rating AS float64)), COUNT(product_id)
2 FROM tkh-project-485015.warehouse_analysis.amazon_products
3 GROUP BY category
4 HAVING AVG(SAFE_CAST(rating AS float64)) > 4.0 AND Count(product_id) > 5
```

This query will process 131.38 KB when run.

Using on-demand processing quota

Query results

Create conversation

Save results

Open in

Job information

Results

Visualization

JSON

Execution details

Execution graph

Row	category	f0_	f1_	
1	Home&Kitchen(Kitchen&Home Appliances)SmallKitchenAppliancesHandBlenders	4.057894736842...		19
2	Electronics(WearableTechnolog...	4.0249999999999...		76
3	Computers&Accessories(Acce ssories&Peripherals)Cables&Accessories(Cables)USBCables	4.151931330472...		233

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